

$\beta\gamma$ ghost algebras and the Verlinde formula

Simon Wood

Cardiff University
School of Mathematics

Joint work with David Ridout

Rational vertex operator algebras

- Module theory is semisimple.
→ Virasoro L_0 operator is diagonalisable.
- Finitely many inequivalent simple modules.
- Span of module characters carries a representation of the modular group $\mathrm{SL}(2, \mathbb{Z}) = \langle T, S \mid S^4 = 1, S^2 = (ST)^3 \rangle$. [Zhu]
- Module category is a *modular tensor category*. [Verlinde, Moore-Seiberg, Huang]

Fusion product derived
from correlation functions

=

Verlinde product derived
from $\mathrm{SL}(2, \mathbb{Z})$ action on
characters

$$M_i \times M_j = \bigoplus_k N_{i,j}^k M_k, \quad N_{i,j}^k = \sum_l \frac{S_{i,l} S_{j,l} \overline{S_{k,l}}}{S_{0,l}}$$

Logarithmic vertex operator algebras

- Module theory is **not** semisimple.
→ Virasoro L_0 operator can have **Jordan blocks**.
- Finitely many inequivalent simple modules, **if the C_2 condition is satisfied**.
- Span of **torus 1-point functions** carries a representation of the modular group $SL(2, \mathbb{Z})$, **if the C_2 condition is satisfied**. [Miyamoto]
- Module category is **not** a *modular tensor category*. Instead just a braided tensor category [Huang-Lepowsky-Zhang] (provided the tensor product exists).

Verlinde formula must fail as characters can't distinguish all modules and modular action does not close on characters.

The affine Lie algebra case

- Determined admissible levels and admissible highest weights for affine Lie algebras. Showed that modular group closes on analytic continuations of characters of admissible modules. [Kac-Wakimoto]
- At non-integral admissible levels Verlinde formula gives negative fusion rules. [Koh-Sorba]

Problems are due to working in the “wrong” category for CFT. Correct category must satisfy:

- Closed under conjugation (contragredients/restricted duals).
- Closed under fusion.
- Modules must admit characters and characters, as formal power series, span a representation of the modular group.

$\beta\gamma$ ghosts have the same features as affine algebras but are easier.

The affine Lie algebra case

- Determined admissible levels and admissible highest weights for affine Lie algebras. Showed that modular group closes on **analytic continuations** of characters of admissible modules. [Kac-Wakimoto]
- At non-integral admissible levels Verlinde formula gives negative fusion rules. [Koh-Sorba]

Problems are due to working in the “wrong” category for CFT. Correct category must satisfy:

- Closed under conjugation (contragredients/restricted duals).
- Closed under fusion.
- Modules must admit characters and characters, as formal power series, span a representation of the modular group.

$\beta\gamma$ ghosts have the same features as affine algebras but are easier.

The affine Lie algebra case

- Determined admissible levels and admissible highest weights for affine Lie algebras. Showed that modular group closes on **analytic continuations** of characters of admissible modules. [Kac-Wakimoto]
- At non-integral admissible levels Verlinde formula gives negative fusion rules. [Koh-Sorba]

Problems are due to working in the “wrong” category for CFT. Correct category must satisfy:

- Closed under conjugation (contragredients/restricted duals).
- Closed under fusion.
- Modules must admit characters and characters, as formal power series, span a representation of the modular group.

$\beta\gamma$ ghosts have the same features as affine algebras but are easier.

The affine Lie algebra case

- Determined admissible levels and admissible highest weights for affine Lie algebras. Showed that modular group closes on **analytic continuations** of characters of admissible modules. [Kac-Wakimoto]
- At non-integral admissible levels Verlinde formula gives negative fusion rules. [Koh-Sorba]

Problems are due to working in the “wrong” category for CFT. Correct category must satisfy:

- Closed under conjugation (contragredients/restricted duals).
- Closed under fusion.
- Modules must admit characters and characters, as formal power series, span a representation of the modular group.

$\beta\gamma$ ghosts have the same features as affine algebras but are easier.